

CHAPTER 4-PROCEDURE FOR COOLING LOAD CALCULATION

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4.1 Collection of Required Data

Before starting the calculation, the required information about the conditioned space is collected.

The following data are generally required:

- Room length, width and height
- Wall, roof and floor details
- Door and window dimensions
- Orientation of the room
- Number of occupants
- Lighting details
- Electrical equipment
- Ventilation and infiltration openings
- Indoor and outdoor design conditions

Accurate data collection helps in obtaining a reliable cooling-load value.

4.2 Measurement of Room and Building Details

The dimensions of the selected room are measured using suitable measuring instruments.

Important measurements

- Length (L)
- Width (W)
- Height (H)
- Door area
- Window area
- Area of other openings

The basic room quantities are calculated as:

$$\text{Floor Area} = \text{Length} \times \text{Width}$$

$$\text{Room Volume} = \text{Length} \times \text{Width} \times \text{Height}$$

These values are used in further cooling-load calculations.

4.3 Determination of Heat Gain

The different sources of heat gain are identified and calculated separately.

A) Heat Gain Through Building Surfaces

Heat transfer through walls, roof, windows and doors is considered.

The heat gain depends on:

- Area of the surface
- Construction/material
- Thermal properties
- Temperature difference
- Solar conditions

For conduction through a building surface, a basic relation is:

$$Q = U \times A \times \Delta T$$

Where:

- Q = Heat transfer (W)
- U = Overall heat-transfer coefficient ($\text{W}/\text{m}^2 \cdot \text{K}$)
- A = Surface area (m^2)
- ΔT = Temperature difference (K)

B) Solar Heat Gain

Solar radiation can produce heat gain through windows and exposed building surfaces.

The amount of solar heat gain depends on:

- Window area
- Glass characteristics
- Solar radiation
- Orientation
- Shading conditions

Solar heat gain through windows is considered separately where applicable.

C) Occupant Heat Gain

People inside the room release heat due to body metabolism.

The calculation depends on:

- Number of occupants
- Activity level
- Sensible heat per person
- Latent heat per person

Therefore, both sensible and latent heat from occupants are considered.

D) Lighting Heat Gain

Electrical lighting produces heat inside the room.

A basic calculation is:

$$Q_l = \text{Number of lamps} \times \text{Power per lamp}$$

Where:

- Q^l = Lighting heat gain
- Power is expressed in watts

E) Equipment Heat Gain

Computers, printers and other electrical equipment release heat during operation.

A basic estimation can be made from the equipment power rating:

$$Q_e = \text{Number of equipment} \times \text{Power per equipment}$$

Actual operating conditions should be considered where possible.

4.4 Ventilation and Infiltration Load

Outdoor air entering the room produces additional cooling load.

Ventilation

Ventilation is the intentional supply of outdoor air.

Infiltration

Infiltration is the uncontrolled entry of outdoor air through:

- Doors
- Windows
- Gaps
- Other openings

The outdoor air contributes both sensible and latent load.

The sensible component can be estimated using:

$$Q_s = \dot{m} \times C_p \times \Delta T$$

Where:

- Q^s = Sensible heat load (W)
- $\dot{m}$ = Mass flow rate of air (kg/s)
- C^p = Specific heat of air (J/kg·K)
- ΔT = Temperature difference (K)

The latent component is related to the difference in moisture content between outdoor and indoor air.

For the HOD cabin, continuously open doors and small open/netted openings are particularly important.

4.5 Calculation of Total Cooling Load

After calculating the individual heat gains, the loads are combined.

The main components are:

1. Wall heat gain
2. Roof heat gain
3. Window heat gain
4. Door heat gain
5. Solar heat gain

4.6 Conversion of Cooling Load and AC Capacity

The calculated total cooling load is converted into a suitable unit for AC selection.

Conversion

$$1 \text{ TR} = 3.517 \text{ kW}$$

Therefore:

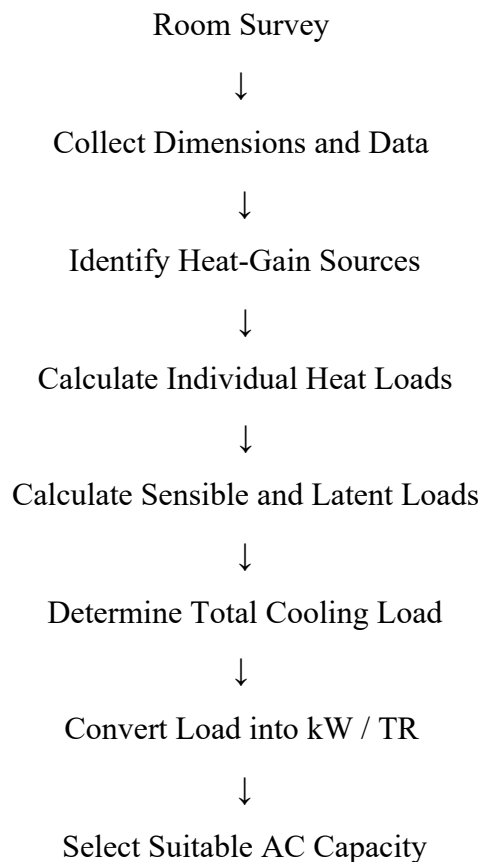
$$\text{Cooling Capacity (TR)} = \text{Total Cooling Load (kW)} / 3.517$$

The calculated value is then compared with available AC capacities.

The selected AC should have sufficient capacity to meet the calculated cooling requirement without unnecessary oversizing.

4.7 Flow of Cooling Load Calculation

The overall procedure can be represented as:



For the present project, this procedure will be applied to the HOD cabin using the actual measured